

Mathieu Celerier

POSTDOCTORAL RESEARCHER

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Summary

Postdoctoral researcher in CNRS-AIST JRL (Joint Robotics Laboratory) at National Institute of Advanced Industrial Science and Technology (AIST) in Tsukuba, Japan, with University of Montpellier (UM), France, through a collaboration between UM and AIST.

My research interest includes semi collaborative physical human-robot interaction and human centered perception of physical human-robot interaction.

I obtained my M.Sc in Microelectronic and Automation Engineering in 2021 from Polytech Montpellier Engineering School. I carried out my final Engineering School Internship as Research Engineer, working on Dual Arm Manipulation of Deformable Objects at Laboratory of Informatics, Robotics and Microelectronics of Montpellier (LIRMM), France.

Work Experience

CNRS-AIST Joint Robotics Laboratory

Tsukuba, Japan

POSTDOCTORAL RESEARCHER

June 2022 - Present

Conduct research on understanding and improving physical human-robot interaction.

Informatic, Robotic and Microelectronic Laboratory of Montpellier (LIRMM)

Montpellier, France

ROBOTICS INTERNSHIP

Feb. 2021 - July 2021

- Assisted PhD student on the subject of dual arm manipulation of deformable objects.
- Development in C++ of robotic applications using the Robot Kinematic Control Library (RKCL) for dual arm manipulation of deformable objects.
- Assisted the PhD student with Python programming for optimization/improvement and for linking Python and C++.

Informatic, Robotic and Microelectronic Laboratory of Montpellier (LIRMM)

Montpellier, France

ROBOTICS INTERNSHIP

May 2020 - Aug. 2020

- V-REP (CoppeliaSim) simulation model of HOAP-3 robot.
- Development in C++ of robotic applications using the Robot Kinematic Control Library (RKCL) for dual arm manipulation of deformable objects.
- Assisted the PhD student with Python programming for optimization/improvement and for linking Python and C++.

Bristol Robotics Laboratory

Bristol, United Kingdom

UNDERGRADUATE INTERNSHIP

June 2019 - Aug. 2019

- Development of a web application implementing scratch programming framework (Blockly) for Anki's Vector robot. The target was to help children discover robotics and programming for the Robotics Week in UK.
- Development of a ROS1 wrapper for Anki's Vector robot API, and implementation of the ROS Navigation Stack on the robot.

Education

University of Montpellier CNRS-AIST Joint Robotics Laboratory

Tsukuba, Japan

PH.D

Sep. 2022 - Jan. 2026

Conduct research on understanding and improving physical human-robot interaction.

University of Quebec in Chicoutimi

Remote from France due to

COVID-19

STUDY SEMESTER

Sep. 2020 - Dec. 2020

Courses on Neural Networks, Quantitative methods for engineers, Engineering project management and Advance numeric communication systems.

Polytech Montpellier

Montpellier, France

MASTER DEGREE IN ENGINEERING

Sep. 2018 - Aug. 2021

- Microelectronics and Automation Major, Specialized in Robotics.
- Smart Bird Feeder project: C programming on STM32, Low-Power constraints, Logging system, Picture capture when a bird arrives and saving on SD card.
- Gravity torque computing for compliance project: Developed Python URDF parser, recursive tree exploration, gravity torque computing with Python/C++.
- Humanoid walk study project: Center of mass trajectory modeling from each step balance point under MATLAB.
- 6 DoF manipulator arm project: Modelling of a manipulator arm using modified Denavit-Hartenberg parameters, dynamic control of the arm, made from scratch using Julia programming language.

IUT of Nîmes

TECHNICAL ASSOCIATE'S DEGREE IN INDUSTRIAL COMPUTING AND ELECTRICAL ENGINEERING

Nîmes, France

Sep. 2016 - Aug. 2018

- Machine learning project to recognize poker cards:
- C++ programming.
- Neural Network based recognition.
- Neural Network algorithms (MLP/CNN) self-taught, made from scratch (Python/MATLAB).